Global Air Pollutaion– Data Analysis

Business Objective

Air pollution poses serious risks to public health, environmental sustainability, and economic productivity. Governments and environmental agencies collect large volumes of air quality data covering pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and AQI across countries, states, cities, and time periods. However, decision-makers often lack clear, actionable insights into pollution drivers, regional disparities, and the effectiveness of mitigation efforts.

This project aims to conduct an **end-to-end analysis of the Global Air Pollution dataset** to:

1. Identify key pollutants driving poor air quality.
2. Understand spatial and temporal variations in air pollution levels.
3. Analyze differences in air quality across countries, regions, and cities.
4. Assess the impact of population exposure and urbanization on air quality.
5. Provide data-driven recommendations to improve air quality management and environmental policy decisions.

Problem Statement

Environmental agencies and policymakers seek to understand the causes and patterns of air pollution to design effective control strategies. Despite having detailed air quality measurements, stakeholders lack clarity on:

- Why air pollution levels vary significantly across regions and countries
- Which pollutants contribute most to unhealthy air quality
- How air quality has changed over time
- Which areas face the highest population exposure risks
- Where mitigation efforts should be prioritized

The goal of this analysis is to examine historical global air pollution data to identify trends, uncover pollution hotspots, and generate actionable insights to support environmental planning, public health protection, and sustainability initiatives.

Phase 1: Business Understanding

Business Questions

- Which countries and cities have the highest average AQI?
- How do pollution levels vary across regions and over time?
- Which pollutants are the primary contributors to poor air quality?
- How does urban population exposure relate to AQI levels?
- Which regions show improvement or deterioration in air quality?

Phase 2: Data Understanding

Data Exploration

Q1. Steps Taken During Data Exploration

1. Reviewed dataset structure and attribute definitions.
2. Identified categorical variables (Country, City, Region, Year).
3. Identified numerical variables (AQI, PM2.5, PM10, NO_2, SO_2, CO).
4. Checked for missing, null, and inconsistent values.
5. Verified data types and validated value ranges.
6. Analyzed distributions of AQI and major pollutants.
7. Examined country-level and city-level pollution patterns.
8. Reviewed pollutant-wise contribution trends.
9. Identified pollution outliers and extreme events.
10. Performed initial aggregations and pivot table analyses.

Q2. Insights Gained During Data Exploration

- Significant variation exists in AQI levels across countries and cities.
- PM2.5 and PM10 are the dominant contributors to poor air quality.
- Urban and industrial regions consistently record higher pollution levels.
- Some countries show gradual improvement, while others exhibit worsening trends.

- Seasonal patterns influence pollution severity in several regions.
- Population-dense cities experience higher exposure risks.
- Regulatory policies appear to impact pollution levels over time.

Q3. Calculated Columns Created

1. **Pollutant Contribution Index**
   Composite score based on PM2.5, PM10, and NO_2 levels
2. **AQI Category**
   Good / Moderate / Unhealthy / Very Unhealthy / Hazardous
3. **Year-on-Year AQI Change**
   Current Year AQI − Previous Year AQI
4. **Urban Exposure Risk Level**
   Low / Medium / High based on AQI and population
5. **City Pollution Intensity**
   Low / Medium / High
6. **Regional Pollution Index**
   Average AQI relative to global average

Q4. KPIs Created and Analysis

Key Performance Indicators (KPIs)

- Total Countries Analyzed
- Total Cities Analyzed
- Average Global AQI
- Average PM2.5 Level
- Percentage of Cities with Unhealthy AQI
- Highest Polluted City AQI
- Regional Pollution Index

Insights from KPIs

- Global average AQI remains above safe limits recommended by health organizations.

- PM2.5 contributes disproportionately to unhealthy air quality.
- A large percentage of cities fall under Moderate to Unhealthy AQI categories.
- Pollution severity varies significantly by region.
- Improvements are uneven across countries.

Phase 3: Data Preparation

Data Cleaning

1. Handling Dimension and Measure Names

- Standardized column naming conventions.
- Removed special characters and inconsistencies.
- Renamed ambiguous pollutant fields for clarity.

2. Handling Missing Values

- Identified missing AQI and pollutant measurements.
- Applied appropriate imputation or removed incomplete records.
- Ensured completeness of geographic identifiers.

3. Data Type Formatting

- Converted categorical variables (Country, City, Region) to text.
- Ensured numerical pollutant values were correctly formatted.
- Verified AQI values were within realistic bounds.

Feature Engineering

- Derived AQI categories for risk classification.
- Calculated pollutant contribution scores.
- Created year-on-year trend indicators.
- Classified cities by pollution intensity.
- Developed regional benchmarking indices.

Data Integration and Formatting

- Ensured consistency across geographic hierarchies.
- Removed duplicate city-year records.
- Standardized country and region names.
- Prepared dataset for analysis and visualization.

Phase 3.1: Data Exploration and Analysis

Analysis Performed

- Country-wise and city-wise AQI comparison.
- Pollutant contribution analysis.
- Temporal trend analysis of AQI and PM2.5.
- Regional pollution hotspot identification.
- Population exposure risk assessment.
- Outlier detection for extreme pollution events.
- Improvement vs deterioration trend analysis.

Key Findings

- Air pollution levels vary significantly across countries and cities.
- PM2.5 is the most critical pollutant impacting global air quality.
- Urban and industrial regions face the highest pollution burden.
- Some regions show improvement due to stricter environmental regulations.
- Population exposure to unhealthy air remains a major global concern.

Recommendations

1. Prioritize reduction of PM2.5 emissions through cleaner energy and transport.
2. Implement stricter air quality regulations in high-risk urban areas.
3. Promote real-time air quality monitoring and public awareness.
4. Target pollution control efforts in identified hotspot regions.
5. Use regional benchmarks to evaluate policy effectiveness.